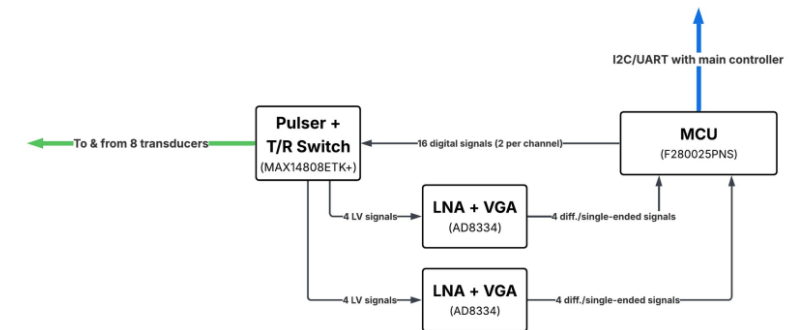
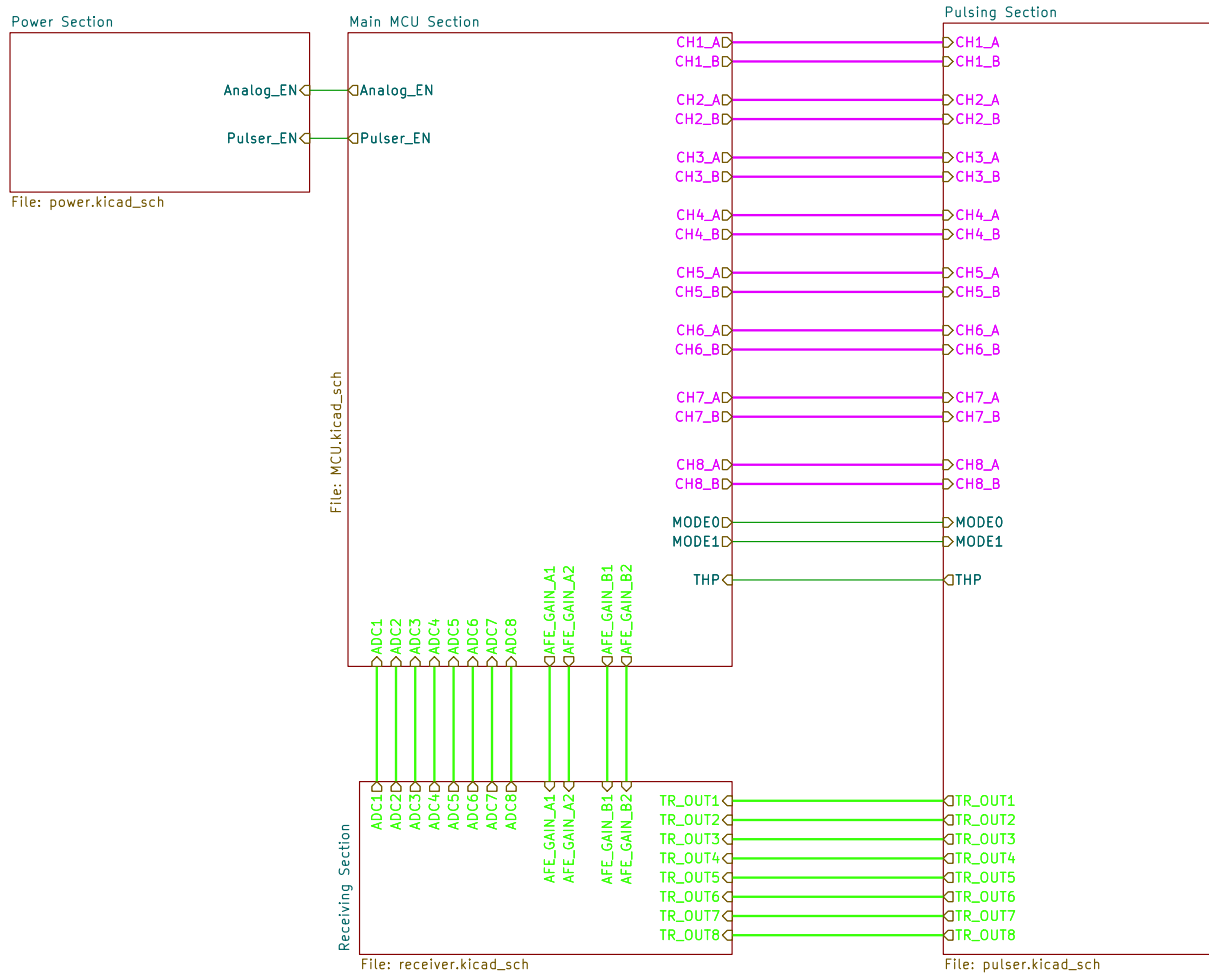
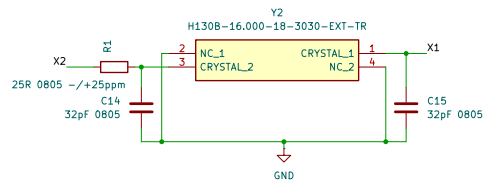




12 Volt (External) (555mA)	7 Volt (Buck Converter) (350mA)	5 Volt (LDO Regulator) (200mA)	LNA + VGA Supply (160mA – short circuit)
		3.3 Volt (LDO Regulator) (150mA)	Op-Amps (4 * 10mA = 40mA)
	5 Volt (Buck Converter) (205mA)	MCU Supply (150mA)	
		Positive Pulse Driver Supply (100mA)	
		3.3 Volt (LDO Regulator) (5mA)	Pulser Logic Supply (5mA)
		-5 Volt (Inverting Buck Converter) (100mA)	Negative Pulse Driver Supply (100mA)



$C1 = C2 = 2(C_{load} - C_{stray}) = 2(18 - 2) = 32pF$
 The crystal is required ($\approx 16MHz$):
 - To have a drive level of $\geq 1mW$ (a damping resistor is used)
 - Have maximum ESR of 75 Ohms (50 is used)
 - $12pF < C1 < 24pF$ (18pF is used)

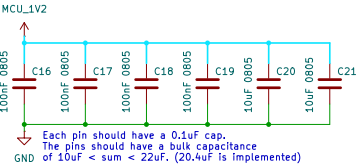
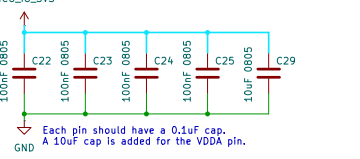
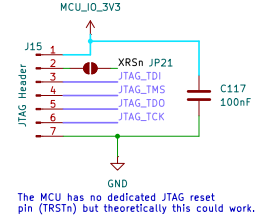
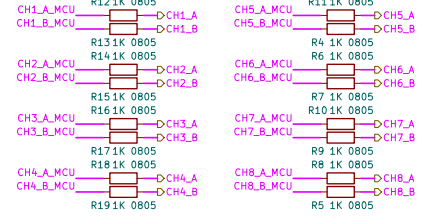
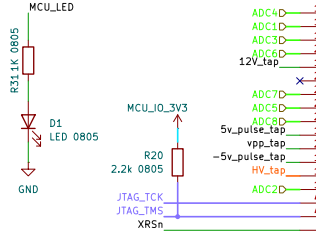
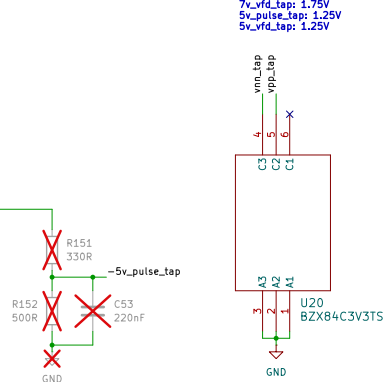
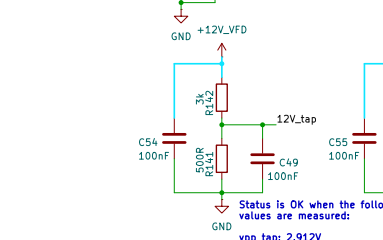
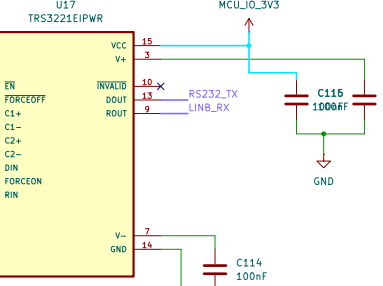
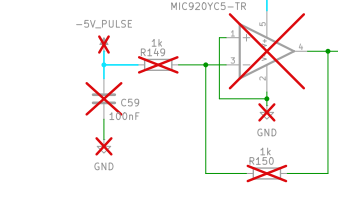
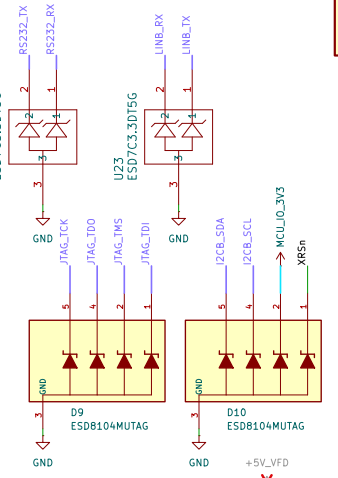
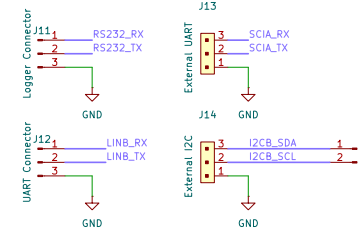
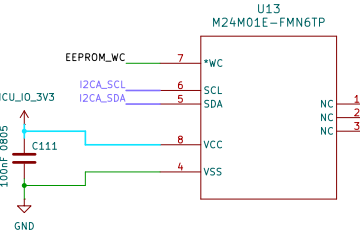
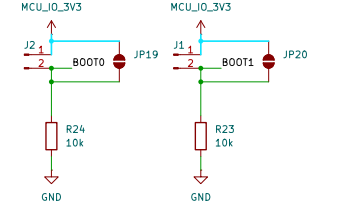
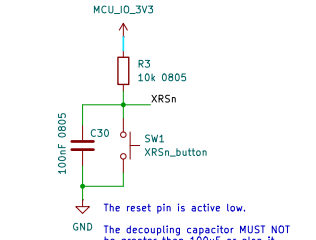


Table 3-1: Device Default Root Modes

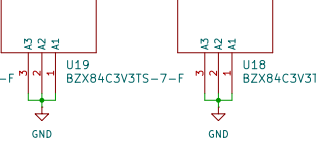
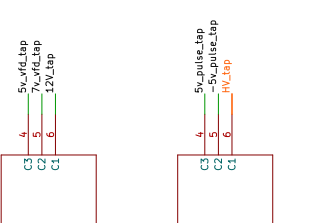
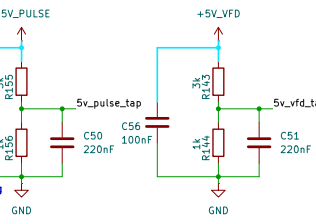
BOOT MODE	OPROOF DEFAULT BOOT MODE SELECT	OPROOF BOOT MODE SELECT	OPROOF DEFAULT BOOT MODE SELECT
PowerID	0	0	0
SCIA_TX	0	1	0
SCIA_RX	1	0	0
Flash	1	1	1



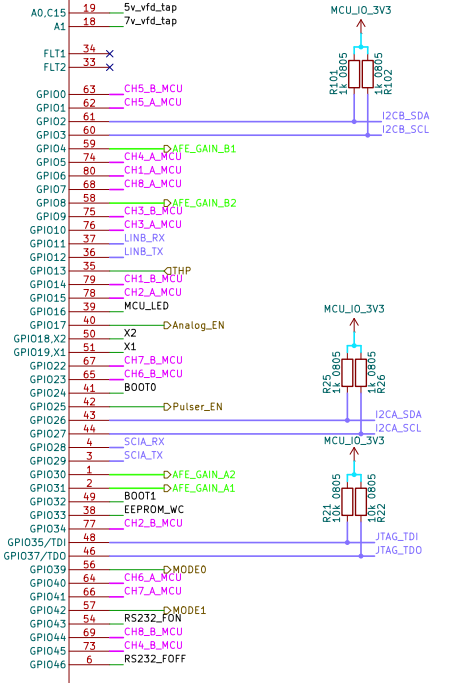
JTAG test mode select (TMS) with internal pullup. This serial control input is clocked into the Tap controller on the rising edge of TCK. This device does not have a TEST pin, an external pulser resistor (recommended 2.2 kΩ) on the TMS pin to VDDIO should be placed on the board to keep JTAG in reset during normal operation.
 The debugger to be used is the XDS110 debugger (recommended by TI).



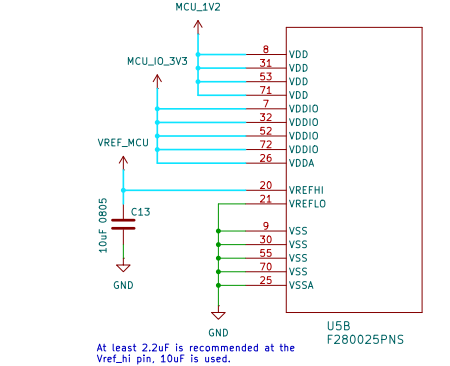
The reset pin is active low.
 The decoupling capacitor MUST NOT be greater than 100uF or else it will interfere with the watchdog reset functionality.



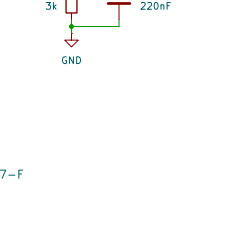
Flash test pins (whatever they are). Must be left unconnected.



1K resistors are placed at pulser lines to limit current to 3.3mA per pin.

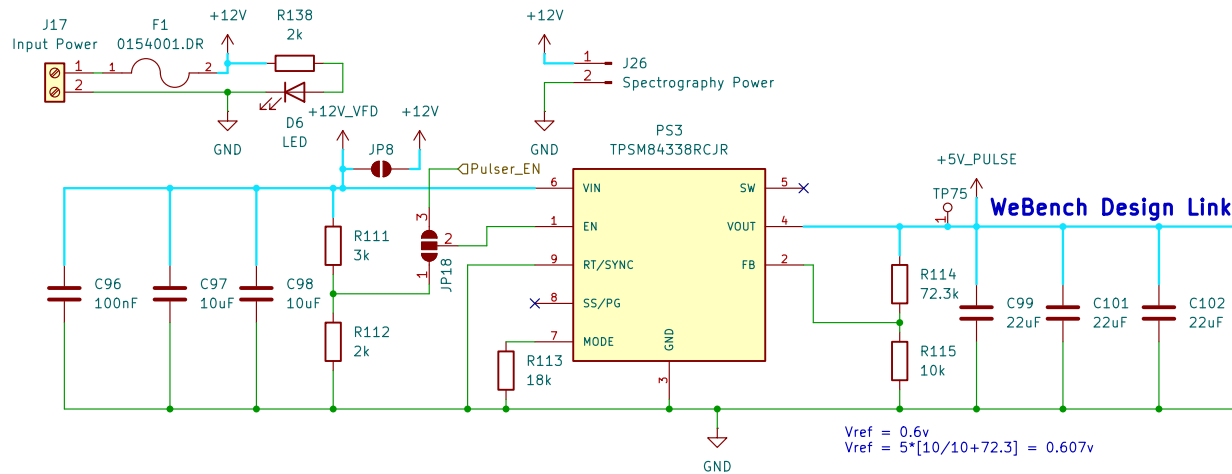
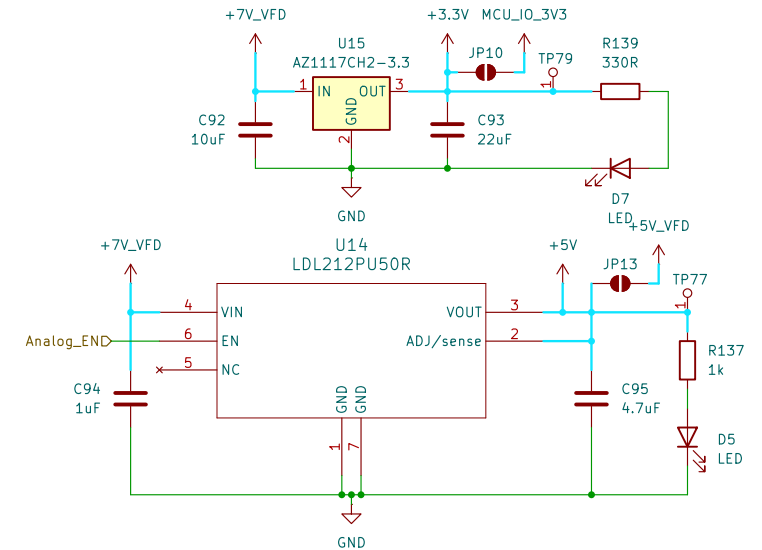
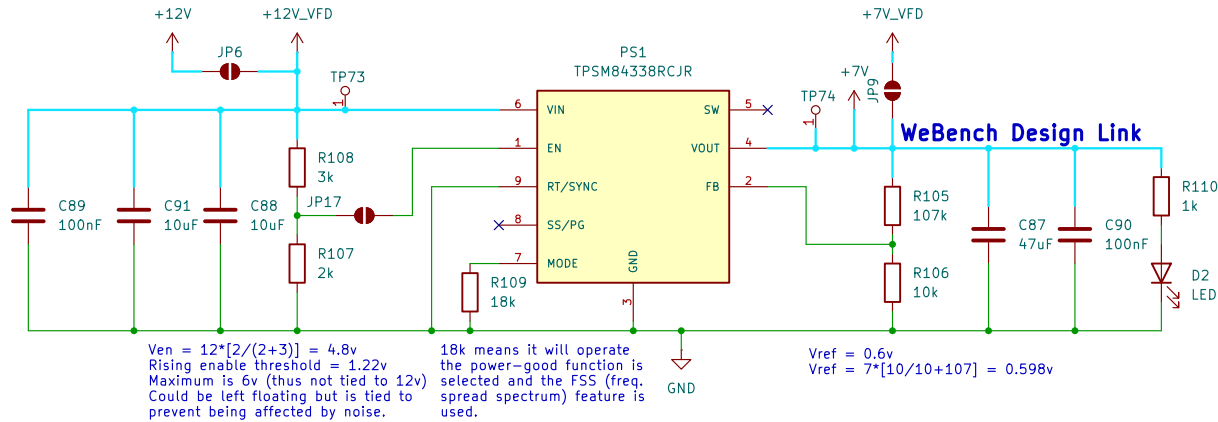


At least 2.2uF is recommended at the Vref_hi pin, 10uF is used.

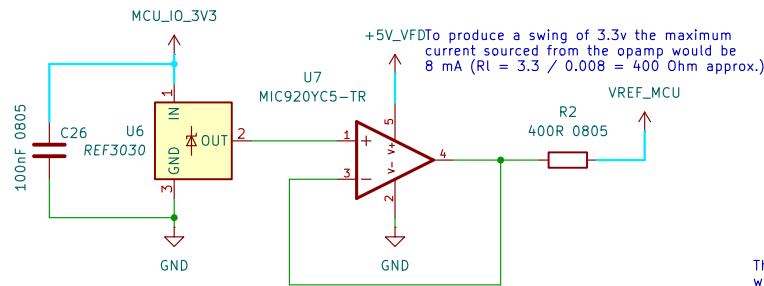
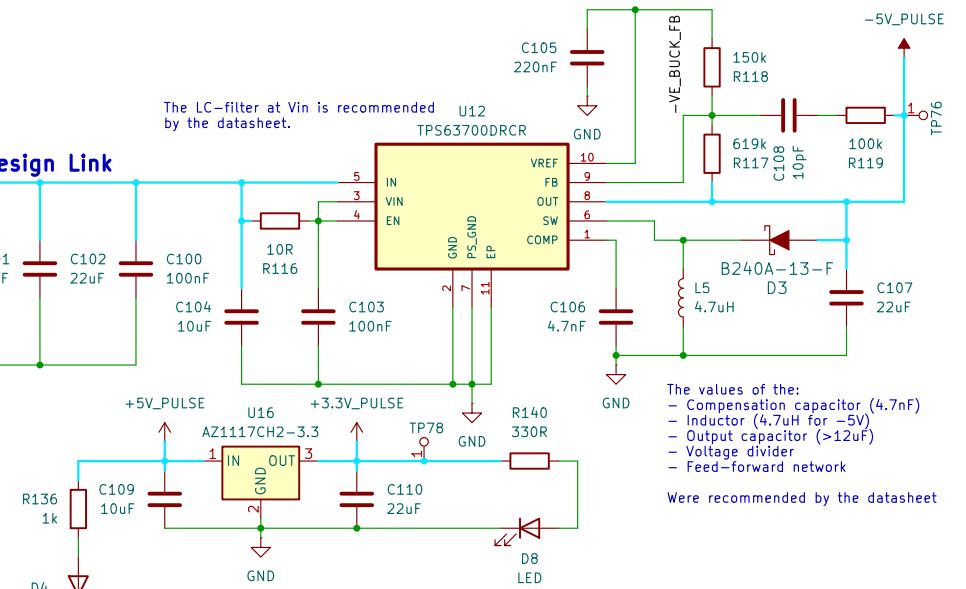


VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.

In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.



The LC-filter at Vin is recommended by the datasheet.



These decoupling capacitor values were recommended by the datasheet